SCOREBOARD – Hadamard 14252 Paley-I (2026-10-09)

Cutoff: 2026-10-09T09:32:06Z (~5:32AM ET)

Object	Status	max|HHˆT-nI|	Wall (s)	SPK
H_14252 Paley-I (q=14251)	PASS	0	2610.9926	**IN SPK** (Paley-I)
H_14252 Paley-II (q=7125)	FAIL/SKIP	–	–	7125=3*5ˆ3*19 NOT prime power; ===1 mod 4 but composite
H_14252 Kronecker (4*3563)	FAIL/SKIP	–	–	3563=7*509 ===3 mod 4 odd NOT Hadamard-admissible

Units: 677/677 * row blocks 112/112 * overall_pass=True * wall_total=2653.0945s
Gap 2052: **OPEN** * clay_closed=false * GATE-04 **LOCKED** * conjecture **OPEN**
IN SPK via Paley-I; Miyamoto digression skipped; exclude Miyamoto-only 8948/8644/8516/...; prior SPK 10224+10212+10164+10160+10152+10140+10112+10104+10100+10068+10040+10008+9968+9932+9924+9908+9888+9884 retained

Next optional: **q2=6961->n=13924 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6961 prime ===1 mod 4; Paley-I q=13923=3ˆ2*71317 NOT prime -> FAIL; Kronecker 13924=4*3481; 3481=59ˆ2 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3481); 13924=2ˆ2*59ˆ2 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=6977->n=13956 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6977 prime ===1 mod 4; Paley-I q=13955=5*2791 NOT prime -> FAIL; Kronecker 13956=4*3489; 3489=3*1163 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3489); 13956=2ˆ2*31163 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=6997->n=13996 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6997 prime ===1 mod 4; Paley-I q=13995=3ˆ2*5*311 NOT prime -> FAIL; Kronecker 13996=4*3499; 3499 odd prime (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3499); 13996=2ˆ2*3499 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7001->n=14004 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7001 prime ===1 mod 4; Paley-I q=14003=11*1967 NOT prime -> FAIL; Kronecker 14004=4*3501; 3501=3*2389 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3501); 14004=2ˆ2*3*2389 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7013->n=14028 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7013 prime ===1 mod 4; Paley-I q=14027=13ˆ2*83 NOT prime -> FAIL; Kronecker 14028=4*3507; 3507=3*7167 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3507); 14028=2ˆ2*37167 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7057->n=14116 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7057 prime ===1 mod 4; Paley-I q=14115=3*5941 NOT prime -> FAIL; Kronecker 14116=4*3529; 3529 odd prime (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3529); 14116=2ˆ2*3529 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7069->n=14140 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7069 prime ===1 mod 4; Paley-I q=14139=3ˆ2*1571 NOT prime -> FAIL; Kronecker 14140=4*3535; 3535=5*7101 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3535); 14140=2ˆ2*5*7101 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7109->n=14220 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7109 prime ===1 mod 4; Paley-I q=14219=59*241 NOT prime -> FAIL; Kronecker 14220=4*3555;

3555=3^2*5*79 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3555); 14220=2^*2^3^2*5*79 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | FIRST Paley-II DIRECT after 14252: q2=7129->n* NOT prime -> FAIL; Kronecker *14260=4*3565; 3565=5*23*31 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3565); 14260=2^2*5*23*31 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII); FIRST any direct route after 14252; comes before the next Paley-I 14304) | Kronecker-only (no direct Paley route for n; note-only, NOT on box): 14288=4*3572 (H_3572 via Paley-I q=3571 prime ===3 mod 4; 14288=2?*19*47) – 14256=4*3564 and 14272=4*3568 are order-admissible but H_m has NO direct classical Paley route (3563=7*509, 1781=13*137; 3567=3*29*41, 1783 prime ===3 mod 4) -> FAIL/SKIP | next Paley-I DIRECT: q=14303->n=14304 (prime ===3 mod 4; no prime ===3 mod 4 in 14252..14302; 14281/14293 ===1 skipped; Paley-II q2=7151 prime but ===3 mod 4 needs ===1 -> FAIL/SKIP; Kronecker 14304=4*3576; 3576=2^3*3*149 ===0 mod 4 order-admissible but H_3576 has NO direct classical Paley route (3575=5^2*11*13 composite; 1787 prime but ===3 mod 4) -> FAIL/SKIP note-only (do NOT invent H_3576); 14304=2?*3*149 ===0 mod 16) (or gap 2052 / Track B)**